

Machine Learning Engineer Technical Interview Case Study

Neutreeno – An Intelligent Decarbonisation Platform

Overview

This take-home case study assesses your ability to leverage out-of-the-box ML tools, work with unstructured emissions data, and deliver practical solutions in a fast-paced startup environment. You'll demonstrate skills directly relevant to Neutreeno's mission of revolutionizing global decarbonisation through AI-driven climate technology.

Timeline

- Spend no more than 2 hours in this task
- Submit 30 minutes before the technical interview

Case Study Context

Neutreeno is building an emission factor matching system that connects unstructured company sustainability reports to our proprietary emissions database. Your challenge is to develop a proof-of-concept pipeline that demonstrates your ability to work with real-world climate data using modern ML tools.

Challenge: Emissions Factor Mapping System

Objective: Build an intelligent system to map purchased items from company procurement data to UK DESNZ emission factors using ML techniques.

Provided Resources:

- **UK DESNZ Emission Factors Database** (Excel)
 - 3,000+ emission factors across categories
 - Scope 1, 2, 3 factors with units (kgCO₂e/unit)
 - Category hierarchies and descriptions
 - SIC codes and activity descriptions
- **Sample Company Purchase Report** (Excel):
 - 6-month procurement data from fictional "TechCorp Ltd"
 - 500+ line items including:
 - Product/service descriptions (varied naming conventions) at 3 levels
 - Quantities and units (mixed formats)
 - Supplier names and categories
 - Purchase amounts in GBP
 - Intentionally messy data: inconsistent naming, abbreviations, typos

Requirements:

1. **Build an intelligent matching system using out-of-the-box ML tools**
 - Implement an effective matching pipeline between purchase descriptions and DESNZ categories
 - Use pre-trained models (Hugging Face, Sentence-BERT, or similar)
 - Handle variations: "Company laptops" → "Computers and IT equipment"
 - Match despite typos, abbreviations, and different taxonomies
2. **Implement uncertainty quantification for matches**
 - Assign uncertainty to each mapping that can be used for later uncertainty propagation techniques
 - Flag ambiguous items requiring human review
 - Provide top-3 potential matches when confidence is low
 - Handle items with no suitable emission factor

Evaluation Criteria:

- Smart use of ML models for semantic matching
- Robustness to real-world data quality issues
- Quality of uncertainty handling and user feedback mechanisms
- Code efficiency and scalability for larger datasets
- Clear communication of approach and results

Submission Requirements

Email Submission:

Submit all the below by responding to the email where you received this document.

1. GitHub repository containing your code
2. Results presentation (.pdf or .pptx)

Virtual Presentation Format:

60 minutes total:

- 10 minutes: Present your solution and key findings
- 10 minutes: Technical deep-dive Q&A
- 40 minutes: Technical questions not related to the case study

Important Notes

1. **Use of AI Tools:** You may use any AI coding assistants (GitHub Copilot, ChatGPT, etc.) but must document their use and be able to explain all code during the presentation
2. **Focus on the presentation:** We ask for the repository to check the results you present but you will not be judged on the clarity of the code

3. **External Libraries:** Prioritize established, well-maintained libraries. Document any trade-offs between custom implementation and out-of-the-box solutions
 4. **Time Management:** This case study is designed to be challenging but completable within the given timeframe. Focus on delivering a working MVP rather than perfect solutions
 5. **Questions:** Email questions to miguel.mancera@neutreeno.com. We'll respond within 4 hours during business hours
 6. **Evaluation Focus:** We're particularly interested in:
 - Your ability to quickly evaluate and implement out-of-the-box solutions
 - Pragmatic decision-making for a startup environment
 - Understanding of ML uncertainty and its business implications
 - Clear communication across technical levels
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Good luck! We're excited to see your innovative approaches to accelerating global decarbonisation through ML.